# Richard Lin

U.S. Citizen

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## EDUCATION

**Temple University** *Philadelphia, PA*
*Bachelor of Science in Electrical and Computer Engineering* *Expected Graduation: May 2028*
Minor: Computer Science | Fundamentals of Physics Certificate | GPA: 3.88/4.00 | Dean's List: F24, S25, F25
**Courses:** Classical Control Systems, Signals & Systems, Processor Systems, Engineering Computation IV (GPU Programming)

## PROFESSIONAL EXPERIENCE

**Undergraduate Research Assistant** *Oct 2025 – Present*
*Temple University – Computer Fusion Lab (CFL)* *Philadelphia, PA*
*Advisor: Dr. Li Bai*

- Administered lab website and systems; ensured security, software maintenance and network reliability
- Built a multi-agent LLM system in Python: a service agent gathers a patron's situation, then a library agent (RAG + LLM-rerank) recommends top 3 best-fit devices from a 400+ item catalog (synced daily), helping people with disabilities
- Trained a Vision-Language-Action (VLA) model with LeRobot to enable a robotic arm to pick up and hand objects for individuals with limited mobility; developed ROS-based robot-control in C++ across multiple robots

**Undergraduate Teaching Assistant** *Aug 2025 – Present*
*Temple University* *Philadelphia, PA*

- Assisted faculty with instruction, hands-on activities, and assignment evaluation; held office hours for student support
- **Intro to Engineering:** Guided students building embedded Micro:bit incubator, testing components (LEDs, fans, sensors)
- **Engineering Problem Solving:** Supported wind turbine calculation/data analysis using Excel, Python (NumPy, Matplotlib)
- **Engineering Computation I:** Taught C/C++/Python/Linux/Bash fundamentals, plus basic algorithms and time complexity

**STEPS Ambassador** *Sep 2024 – Present*
*Temple University – Sustainable Temple Energy and Power Scholars (STEPS)* *Philadelphia, PA*

- Mentored STEPS underclassmen through the high-school-to-college transition, providing academic guidance
- Designed and led renewable-energy STEM demos for high schoolers, emphasizing sustainability and real-world engineering

## PROJECTS

**Musical Light Show – ECE 2352: Circuits and Electronics II** *Apr 2026 – May 2026*

- Designed and built an analog audio-processing system that combines left/right audio channels, and separates high and low frequency signals to drive an audio-reactive LED light show
- Implemented dual op-amp stages for summing and volume/gain control using a feedback potentiometer, high/low-pass filters, peak detectors, BJT LED driver circuits
- Calculated values from gains and cutoff frequency using Ohm's Law and KVL; verified in Multisim and hardware within 1%

**Detective Conan AI Glasses (Childhood Dream Project)** *Aug 2026 – Present*

- Architecting smart glasses on Pi Zero 2 W with camera, mic, solar charging, 3D-printed enclosure; Google Gemini API
- Developing software architecture with intent, LLM routing, object detection, OCR, face recognition, and cybersecurity

## CLUBS & LEADERSHIP

**President** *Sep 2024 – Present*
*Institute of Electrical and Electronics Engineers (IEEE) – Temple University* *Philadelphia, PA*

- Led 14-member IEEE executive board in planning outreach programs, technical workshops, and industry networking events, while coordinating with four affiliated chapters and societies (WIE, PES, EMBS, HKN)
- Grew IEEE SAC participation by 50% (14 → 21) through targeted event promotion and outreach during 2025–2026
- Founded the first annual multi-university hardware hackathon with Drexel and UPenn IEEE branches

**Secretary & Treasurer** *Jun 2026 – Present*
*Eta Kappa Nu (IEEE-HKN), Iota Sigma Chapter – Temple University* *Philadelphia, PA*

## SKILLS

**Programming:** Python, C, C++, Java, MATLAB, Bash, SystemVerilog, JavaScript, HTML/CSS, CUDA, AVR Assembly
**AI & Robotics:** Machine Learning, PyTorch, LLM Integration, RAG, VLA, Edge AI, Computer Vision (OpenCV, YOLO), ROS
**Hardware & Tools:** Git/GitHub, Linux, Raspberry Pi, Arduino, FPGA, Circuit Design & Analysis, Soldering, n8n, PCB Design
**Languages:** English, Mandarin Chinese, Fuzhounese, Spanish